

FORMULE DE TAYLOR-YOUNG

Si f est de classe $\mathcal{C}^n$ au voisinage de 0 :

$$f(x) \underset{x \rightarrow 0}{=} f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + o(x^n)$$

PUISSANCES

$$\begin{aligned}(1+x)^\alpha &\underset{x \rightarrow 0}{=} 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots \\ &\quad + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + o(x^n) \\ \frac{1}{1-x} &\underset{x \rightarrow 0}{=} 1 + x + x^2 + \cdots + x^n + o(x^n) \\ \frac{1}{1+x} &\underset{x \rightarrow 0}{=} 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)\end{aligned}$$

EXPONENTIELLE ET LOGARITHME

$$\begin{aligned}e^x &\underset{x \rightarrow 0}{=} 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n) \\ \ln(1+x) &\underset{x \rightarrow 0}{=} x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n+1} \frac{x^n}{n} + o(x^n) \\ \ln(1-x) &\underset{x \rightarrow 0}{=} -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots - \frac{x^n}{n} + o(x^n)\end{aligned}$$

TRIGONOMETRIE

$$\begin{aligned}\sin(x) &\underset{x \rightarrow 0}{=} x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + o(x^{2k+2}) \\ \cos(x) &\underset{x \rightarrow 0}{=} 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^k \frac{x^{2k}}{(2k)!} + o(x^{2k+1}) \\ \tan(x) &\underset{x \rightarrow 0}{=} x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^6) \\ \arctan(x) &\underset{x \rightarrow 0}{=} x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^k \frac{x^{2k+1}}{2k+1} + o(x^{2k+2})\end{aligned}$$

HYPERBOLIQUES

$$\begin{aligned}\operatorname{sh}(x) &\underset{x \rightarrow 0}{=} x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2k+1}}{(2k+1)!} + o(x^{2k+2}) \\ \operatorname{ch}(x) &\underset{x \rightarrow 0}{=} 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2k}}{(2k)!} + o(x^{2k+1})\end{aligned}$$

EXEMPLE DE MANIPULATION

Avec $\alpha = -2$ dans $(1+x)^\alpha$, puis x remplacé par $-x$:

$$\frac{1}{(1-x)^2} \underset{x \rightarrow 0}{=} 1 + 2x + 3x^2 + \cdots + (n+1)x^n + o(x^n)$$

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